

ni

Notational Intelligence as Runtime Commit Law

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board_synth_claimed=false — software reference stack; surrogate joules ≠ board watts.

System One decides in software. WCA decides whether the machine is allowed to move.

A comprehensive research study for OpenIE / WCA

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Honesty	No fabricated citations. Claims carry real URLs / arXiv / DOI, or are marked [UNVERIFIED] / [VENDOR] / [SURROGATE] . Stack is software reference: board_synth_claimed=false. Surrogate OpCounter joules ≠ board watts.

Abstract

Linus Lee's *notational intelligence* (thesephist, 2022) argues that inventing better notations can raise effective intelligence more than automated tools alone. That claim is historically fertile and intellectually generous. This study treats Lee as **Brahe-scale**: precise observation that notation amplifies thought, without yet stating the dynamical law that agentic Physical AI must obey at the moment of irreversible action.

David Charlot's program—**WCA / EFA Commit Gate** in `wca-lut-edge`—is framed as the **Keplerian step**: notation as **runtime commit/refuse law**. The product nomenclature is versioned wire types under `wca.commit.v1`:

Proposal → **Certificate** → **RefuseReason** → **CommitDecision**

Composition is conjunctive and explicit:

```
commit = lut_allow ∧ energy_ok [∧ cbf_ok]
```

The stack complements **System One** (typed propose / decision tax collapse), does **not** compete with Taalas / Groq / Etched (silicon tokens/J) or World Labs—class world models (predict-then-act), and meters energy under OpenIE-style **joule honesty** with labeled tiers. Rust 1.98 (crates/wca-commit) is the hot path; FPGA board synthesis is explicitly **not** claimed.

Primary finding. Lee is right that notation is under-invested. Charlot is right that for agentic machines the expensive mistakes are *commits* (joules, torque, irreversible tools), so the notation that matters is the one the runtime cannot ignore. Sep 29–30 2026 landscape scans did **not** find a published product stack that ships Boolean allow-LUT ∧ port-Hamiltonian / Safe energy certificate ∧ cheap TLMM (or System One) proposal as a single auditable commit boundary (S0TA_SCAN.md §1.8; PRODUCT_LANDSCAPE.md §3.3). That white space is real; **silicon joule leadership is not**.

0. Method

Date & timezone. Research and drafting occurred **Tue Sep 29 – Wed Sep 30, 2026**, America/New_York (EDT, UTC−4). Box-local date confirmed EDT during verification runs.

Sources.

1. **Primary essay:** WebFetch of <https://thesephist.com/posts/notation/> (Lee, 2022-01-02).
2. **Secondary NI / TtT:** Gradient conversation; thesephist speaking index (Compile / Cursor talk listed Jun 2026); Matuschak & Nielsen TtT essay; Iverson / Engelbart / Kay / Victor primary URLs.

- 3. **Formal methods & safety:** Ames CBF; Alshiekh shielding; Dawson learned certificates; Manek & Kolter stable dynamics; Roth stable pH-NN; Leung & Paré energy-aware Bayesian CBF; Yu / Žikelić / Henzinger certificate repair via runtime monitoring; Sánchez et al. RV survey; Falcone / Bartocci RV intros.
- 4. **Energy / LUT / edge:** DiffLogic, DWN, BitNet, T-MAC, TeLLMe, Hooker Hardware Lottery, Landauer, Horowitz ISSCC, Sandberg brain energetics.
- 5. **Agents / tools:** Anthropic MCP announcement; OSWorld-Human; AutoGen; MetaGPT.
- 6. **Local stack:** README demos executed on the box (wca-soc-loop seed-1 golden; wca-mcp-gate --demo); schemas under wca.commit.v1 (research archive); Rust sources under crates/wca-commit (research archive).

Honesty protocol. Every external factual claim must either (a) cite a live URL / arXiv / DOI checked in this window, or (b) carry an honesty tag. Vendor throughput / watt claims stay **[VENDOR]**. OpCounter × analytical E_* stays **[SURROGATE]**. Missing literature is labeled **[UNVERIFIED]** rather than invented.

Non-goals. This study does not claim board watts, brain-matching efficiency, or “first energy-aware AI.” It does not punch down on Lee. Star Trek mappings are metaphor only.

1. Executive synthesis (Brahe → Kepler)

1.1 Thesis

Layer	Lee / Tft lineage	Charlot WCA / EFA
What notation is	Medium that amplifies human thought (Iverson, Engelbart, Kay, Victor, Lee)	Executable commit nomenclature wired into SoC / agent tool boundary
Who it primarily serves	Human intellect first; machines as co-authors of notation	Plant / irreversible tool / energy budget first; humans as policy + audit layer
Success metric	“Previously unthinkable thoughts”; mnemonic / creative leverage	False-allow → 0; util/J with refuse reasons; surrogate→DUT joule path
Failure mode	Beautiful demos that don’t change civilization’s thought patterns (Matuschak–Nielsen warning)	Soft confidence strings sold as safety; surrogate J marketed as silicon W

Brahe metaphor (explicit). Tycho Brahe assembled precise observations without a correct dynamical law; Kepler turned data into enforceable planetary laws. Lee assembles the *observation* that notation amplifies intelligence. Charlot’s bet is that **Proposal / Certificate / RefuseReason / CommitDecision** plus EFA energy checks are the *laws* agents must obey at runtime—not optional documentation.

1.2 One-paragraph verdict

Lee is right that notation is under-invested relative to “more automation.” Charlot is right that *agentic* Physical AI fails when notation stops at the whiteboard: LLMs propose in natural language; CBFs and Lyapunovs live in papers; LUTs burn on FPGAs; energy is a blog slide. **WCA’s compositional claim** —Boolean allow-LUT $\wedge$ port-Hamiltonian / Safe energy certificate $\wedge$ cheap TLMM (or System One / agent) proposal as a single commit boundary—was **not found as a published product stack** in the Sep 29–30, 2026 scans. That white space is real; **silicon joule leadership is not**. Treat Lee as intellectual ancestor and Brahe; treat WCA as the Keplerian *law* only when certificates are auditable and joules are labeled by measurement tier.

1.3 Product lock (do not drift)

- Beat Lee on **functional proof**: codebase + whitepaper pathway + Medium — not better chalkboards.
- Compose with System One; do not clone it.
- Do not compete with Taalas / World Labs for silicon or world-model primacy.
- Honesty: board_synth_claimed=false; OpenIE joule honesty; Rust 1.98 hot path.

2. Literature: notational intelligence lineage (deepened)

2.1 Classical nodes

Node	Year	Primary URL	One-line claim	Relation to Charlot
Kenneth E. Iverson , <i>Notation as a Tool of Thought</i> (Turing Award; CACM)	1979/1980	https://dl.acm.org/doi/10.1145/358896.358899 ; PDF https://www.eecg.utoronto.ca/~jzhu/csc326/readings/iverson.pdf	Executable + unambiguous programming notation as a tool of thought (APL).	Closest classical ancestor to “notation that runs.” WCA is Iverson for <i>commit</i> , not array ops.
Douglas Engelbart , <i>Augmenting Human Intellect</i>	1962	https://www.dougelbart.org/pubs/augment-3906-Framework.html	H-LAM/T: Human using Language, Artifacts, Methodology, Training as one augmented system.	Schemas + gate + human policy layer.
Alan Kay , <i>A Personal Computer for Children of All</i>	1972	https://worrydream.com/refs/Kay_1972_-_A_Personal_Computer_for_Children_of_All_Ages.pdf	Personal dynamic medium that	Medium thesis; Charlot specializes to AI?Machine commit .

Node	Year	Primary URL	One-line claim	Relation to Charlot
<i>Ages</i> (Dynabook)			changes thought patterns.	
Kay & Goldberg, <i>Personal Dynamic Media</i>	1977	Overview https://en.wikipedia.org/wiki/Dynabook	Dynabook vision as personal dynamic media.	UI/medium, not plant certificates.
Bret Victor, <i>Inventing on Principle</i>	2012	https://www.youtube.com/watch?v=PUv66718DII ; archive https://archive.org/details/vimeo-36579366	Immediate connection between idea and consequence.	Motivates short propose→certify→sens loops.
Bret Victor, <i>Media for Thinking the Unthinkable</i>	2013	https://worrydream.com/MediaForThinkingTheUnthinkable/	Representations must upgrade for unthinkable system understanding.	Supports commit nomenclature as representation upgrade
Linus Lee (thesephist), <i>Notational intelligence</i>	2022-01-02	https://thesephist.com/posts/notation/	Better notations can contribute <i>more</i> than automated tools to effective intelligence; coins notational intelligence .	Brahe node. Dynamic/computable notation; human-first; no refuse-to-commit law.
Linus Lee x The Gradient, <i>At the Boundary of Machine and Mind</i>	2022	https://thegradientpub.substack.com/p/linus-lee-at-the-boundary-of-machine	Notation vs language; inventing notations; LM interfaces.	Public expansion; still interface/creative-work centered.
Linus Lee, <i>Notational Intelligence</i> talk (Compile / Cursor)	listed Jun 2026	https://thesephist.com/ ; YouTube index https://www.youtube.com/watch?v=rv_VS189aVI	Mechanizing invention of new alphabets/ notations with DL.	Meta-notation R&D; orthogonal to plant energy certificates. Talk details beyond listing are [UNVERIFIED] without transcript audit.
Andy Matuschak & Michael Nielsen, <i>How can we develop transformative</i>	2019	https://numinous productions/ttft/	TtT undersupplied; insight-through-making; mnemonic medium.	Diagnostic of why NI stalls commercially.

Node	Year	Primary URL	One-line claim	Relation to Charlot
<i>tools for thought?</i>				
François Chollet , <i>On the Measure of Intelligence</i>	2019	https://arxiv.org/abs/1911.01547	Intelligence $\approx$ skill-acquisition efficiency / generalization.	Lee cites Chollet; Charlot adds an energy denominator .
Terence Tao (notation properties; via Lee)	various	Summarized in Lee's essay	Unambiguity / expressiveness / suggestiveness / natural transformation.	Apply checklist to <code>wca.commit.v1</code> .

Lineage judgment. Iverson $\rightarrow$ Engelbart $\rightarrow$ Kay $\rightarrow$ Victor $\rightarrow$ Lee $\rightarrow$ Matuschak/Nielsen: **symbols + media change what thoughts are cheap**. Charlot's discontinuity: for agentic Physical AI, expensive mistakes are commits, so the notation that matters is the one the runtime cannot ignore.

2.2 What Lee actually says (fair paraphrase)

From the live fetch of the 2022 essay:

- Inventing better notations can contribute **far more** than automated tools to effective intelligence.
- Good notation properties (via Tao): unambiguity, expressiveness, suggestiveness, natural transformation.
- **Computable notation** (programming languages, Wolfram Language, interactive media) is a new class: notation that executes.
- **Dynamic notation** (Victor-aligned): software should make symbols interactive, not clone paper into pixels.
- Success = enabling **previously unthinkable** thoughts—not mere productivity.

Charlot **keeps all of this**. The Keplerian move is to specialize “computable / dynamic notation” to the **irreversible boundary**: the symbols are Proposal, Certificate, RefuseReason, CommitDecision; the interaction is allow vs hold; the previously-unthinkable move for agents is **silent unsafe success becoming unrepresentable**.

2.3 Formal methods, runtime verification, certificates

NI literature rarely cites runtime verification (RV). Physical AI commit law *is* an RV-shaped problem: monitors over traces of propose/certify/act.

Work	URL	Why it matters for WCA
Sánchez et al., <i>Survey of Challenges for</i>	https://arxiv.org/abs/1811.06740	RV beyond software; CPS / hybrid domains.

Work	URL	Why it matters for WCA
<i>RV from Advanced Domains</i>		
Bartocci & Falcone (eds.), <i>Lectures on Runtime Verification</i>	https://link.springer.com/book/10.1007/978-3-319-75632-5	Intro + CPS monitoring chapters.
Yu, Žikelić, Henzinger, <i>Neural Control and Certificate Repair via Runtime Monitoring</i>	https://arxiv.org/abs/2412.12996 ; AAAI https://ojs.aaai.org/index.php/AAAI/article/view/34840	Runtime monitor detects certificate / property violations; repair loop. Closest “certificate + runtime” cousin.
<i>Formal Verification of Neural Certificates Done Dynamically</i>	https://arxiv.org/pdf/2507.11987	Lightweight runtime verification of neural certificates over lookahead regions.
Scalable verification of neural CBFs (LBP)	https://arxiv.org/html/2511.06341	Verification bottleneck for neural barriers.

Implication. WCA’s gate is a **narrow, plant-facing monitor** with typed refuse codes—not full LTL monitor synthesis, not SMT-backed neural CBF verification. Kinship is real; scope difference is honest.

2.4 Control certificates: CBF, Lyapunov, port-Hamiltonian

Work	URL	Role
Ames et al., CBF theory & applications (ECC)	https://doi.org/10.23919/ECC.2019.8796030	Set invariance via barrier inequalities.
Alshiekh et al., Safe RL via Shielding	https://arxiv.org/abs/1708.08611	Discrete shields / minimally invasive filters.
Dawson, Gao, Fan, Safe Control With Learned Certificates	https://arxiv.org/abs/2202.11762	Survey of learned Lyapunov / barrier certificates.
Manek & Kolter, Learning Stable Deep Dynamics	https://arxiv.org/abs/2001.06116	Joint dynamics + Lyapunov architecture.
		Physics-informed energy structure.

Work	URL	Role
Greydanus et al., Hamiltonian Neural Networks	https://arxiv.org/abs/1906.01563	
Roth et al., Stable Port-Hamiltonian Neural Networks	https://arxiv.org/abs/2502.02480	pH structure as certificate language.
Zhong et al., Neural Energy Casimir Control	https://arxiv.org/abs/2112.03339	Energy-Casimir control with neural models.
Leung & Paré, Energy-Aware Bayesian CBFs	https://arxiv.org/abs/2512.24493	Energy barriers + GP posteriors; high-probability safety.

WCA stance. Classic SoC uses structural energy ($\dot{V}$ / fixed-point PH) $\wedge$ Boolean LUT—not a full CBF-QP. Dual compose `lut_and_safe_and_cbf` exists in `wca-lyap-plant` / MCP gate default. Gap vs Ames on rich plants is acknowledged; composition is the answer, not reinvention theater.

2.5 Energy-aware ML, LUT compute, hardware lottery

Work	URL	Role
Hooker, <i>The Hardware Lottery</i>	https://arxiv.org/abs/2009.06489	Ideas win when hardware fits them.
Petersen et al., DiffLogic	https://arxiv.org/abs/2210.08277	Differentiable logic $\rightarrow$ LUT/gate nets.
Bacellar et al., DWN	https://arxiv.org/abs/2410.11112	FPGA-oriented weightless / LUT nets.
Ma et al., BitNet b1.58	https://arxiv.org/abs/2402.17764	Ternary LLM era; proposal-path kinship.
Wei et al., T-MAC	https://arxiv.org/abs/2407.00088	CPU table-lookup matmul; energy wins [paper] .
TeLLMe (edge FPGA ternary LLM family)	https://arxiv.org/html/2510.15926v2 ; https://arxiv.org/abs/2504.16266	Board-context figures—gold standard WCA must eventually match.
Landauer 1961	https://www.dna.caltech.edu/courses/cs191/paperscs191/landauer1961.pdf	kT ln 2 floor.
Horowitz, Computing's Energy Problem (ISSCC 2014)	https://gvern.net/doc/cs/hardware/2014-horowitz-2.pdf	pJ-scale op tables for calibrating E_* .

2.6 MCP, agent tool safety, multi-agent

Work	URL	Role
Anthropic MCP announcement		

Work	URL	Role
	https://www.anthropic.com/news/model-context-protocol	Tool/data transport standard (Nov 2024).
OSWorld-Human	https://arxiv.org/abs/2506.16042	CUA tax / computer-use benchmarks.
AutoGen	https://arxiv.org/abs/2308.08155	Multi-agent frameworks.
MetaGPT	https://arxiv.org/abs/2308.00352	Multi-agent software generation.
LeanDojo	https://leandojo.org/ ; https://arxiv.org/abs/2306.15626	Offline theorem proving—different certificate class.

Compose rule. MCP discovers and ships tool calls; WCA owns **irreversible** branches (mcp_gate.rs, wca-mcp-gate). Multi-agent debate does not create plant safety: many propose, **one** commit gate.

2.7 World models, specialized silicon, biology, Active Inference

Covered in pathway map (§4). Primary cites retained: Ha & Schmidhuber World Models (<https://arxiv.org/abs/1803.10122>); DreamerV3 (<https://arxiv.org/abs/2301.04104>); Meta/LeCun JEPA discussion (<https://ai.meta.com/blog/yann-lecun-advances-in-ai-research/>); Groq / Etched / Taalas **[VENDOR]**; Loihi 2 brief; Sandberg (<https://arxiv.org/abs/1602.04019>); cortical partitioning nuance (<https://arxiv.org/abs/2102.06273>); Friston FEP (<https://www.nature.com/articles/nrn2787>); OpenIE joule-lang (<https://github.com/openIE-dev/joule-lang>).

3. Steelmanned counters (expanded)

Each counter is stated in its strongest form, then answered. Keep all eight from the memo; add three from deeper literature.

3.1 Scale-only (“just train bigger”)

Steelman. Scaling lore: capability emerges from scale + data; notation design is boutique.

Reply. Hooker’s Hardware Lottery shows ideas win when hardware fits. DiffLogic / DWN / LogicNets / TeLLMe / T-MAC win by training for LUTs and table-lookup. Scale without a commit law still pays propose→discard and false-act tax. **Compose:** scaled proposers *behind* the gate.

3.2 Sapir–Whorf skepticism

Steelman. Strong linguistic determinism is rejected (McWhorter *The Language Hoax*; Pullum overview <https://pullum.ppls.ed.ac.uk/LingRelativity.pdf>).

Reply. Charlot needs Iverson's weaker claim: executable notation changes which operations are cheap and which errors are expressible. A `RefuseReason` enum does not cage cognition; it makes **silent unsafe commit** unrepresentable as success.

3.3 Natural language is sufficient (LLMs + MCP)

Steelman. LLMs + MCP are the universal interface; typed System One decisions shrink the need for new notation.

Reply. NL + MCP solve discovery and transport, not plant passivity / energy / Boolean allow. System One collapses generation tax; it does not certify Lyapunov / energy before actuators. Probability routes; certificates commit. **Compose:** System One / MCP upstream; WCA on irreversible branches.

3.4 Formalization bureaucracy

Steelman. Certificates and schemas slow iteration; vibe coding ships.

Reply. Bureaucracy is real when formalization wraps everything. WCA formalizes **only the commit boundary** (4-bit allow LUT + PH check; schemas in `artifacts/schemas/wca.commit.v1/`). Read-only / reversible tools may bypass. Ames-style minimally invasive filtering—not full verification of the proposer.

3.5 Verification energy tax

Steelman. Checking certificates can dominate edge budgets.

Reply. Valid for heavy QP / SMT each step. Invalid for burned LUT-6 + fixed-point PH sized for the plant. Seed-1 golden: `lut_reads=16, ph_muls=64`, episode J $\approx 8.6795e-10$ **[SURROGATE]**. Obligation: meter gate J vs proposal J; kill designs where gate > benefit.

3.6 Certificate false security

Steelman. “Certified” becomes cargo-cult; wrong V with a green light is worse than an honest soft policy.

Reply. Strongest objection. Mitigations: continuous twin; Safe Q16.16 `false_allow=0` goal; refuse reason split; never market soft confidence as certificate. Still missing vs SOTA: disturbance ISS, learned verified Lyapunov, Bayesian credibility budgets (Leung & Paré). Document residual assumptions.

3.7 CBF / Lyapunov already solved it

Steelman. Academic CBF-QP and neural Lyapunov make WCA redundant.

Reply. CBF certifies set invariance; Lyapunov/energy certifies decrease / passivity. WCA's $\dot{V}$ gate is structural energy, not a full CBF QP. Composition `lut  $\wedge$  energy_ok  $\wedge$  cbf_ok` is the honest next step already sketched in product docs.

3.8 Surrogate vs measured joules

Steelman. OpCounter × analytical E_{*} is cosplay; MLPerf Tiny / Joulescope / TeLLMe board watts are the adult table.

Reply. Agree. Label board_synth_claimed=false and publish tier A/B/C. Do not put surrogate J in a “watts” claim on Medium.

3.9 Runtime monitoring already covers certificates (NEW)

Steelman. Yu / Žikelić / Henzinger and dynamic neural-certificate monitors already do runtime certificate checking; WCA reinvents RV.

Reply. Those works target neural policy + certificate *repair* and lookahead verification—valuable and citeable. WCA’s contribution is **product nomenclature + LUT-exportable discrete shield + energy conjunct + MCP irreversible beachhead** as a shipping software reference, not a new RV theory paper. Compose: adopt their repair/monitor ideas for learned certificates; keep Boolean LUT for FPGA export.

3.10 World models make commit law unnecessary (NEW)

Steelman. If the world model is accurate enough, predicted unsafe acts are never proposed.

Reply. Prediction error is not zero. Holodeck safety protocols exist in fiction precisely because generators are unbounded. World models / JEPA compose as **proposer imagination**; the gate still refuses physically illegal commits. Physics-informed (HNN / pH-NN) world models compose especially well with EFA certificates.

3.11 Specialized silicon makes software gates irrelevant (NEW)

Steelman. Taalas / Groq / Etched collapse tokens/J; software commit gates are latency theater.

Reply. Tokens/J is the proposal path. It does not own refuse-to-commit for actuators or irreversible tools. Specialized inference lowers proposal cost; Charlot still agrees with the lottery diagnosis → LUT-first, not another HBM race. Vendor tok/s and W remain **[VENDOR]** until independent DUT.

4. Pathway map vs WCA / OpenIE / System One

4.1 Compete vs compose table

Pathway	Stance if claimed as substitute	Compose with WCA	Key cites
World models	“Prediction is safety”	Proposer / imagination	Ha & Schmidhuber; DreamerV3; JEPA blog

Pathway	Stance if claimed as substitute	Compose with WCA	Key cites
Etched / Groq / Taalas	"Tokens/J is the whole game"	Cheap proposal path	Vendor sites [VENDOR]; Hooker
System One (Jev / Laya)	"Typed NL replaces commit law"	Upstream decision tax cut	https://laya-ai.com/system-one-models ; arXiv 2503.23303, 2510.01237
Lean / neuro-symbolic	"Proofs replace runtime gate"	Offline properties	LeanDojo
MCP / CUA agents	"Tools are enough"	Transport + irreversible class	Anthropic MCP; OSWorld-Human
CBF / shields	"Already solved"	Add cbf_ok conjunct	Ames; Alshiekh; Dawson
pH / Landauer / thermodynamic	"Theory without product"	EFA certificate language	Roth; Landauer; OpenIE joule-lang
Neuromorphic	"Spikes replace LUTs"	Sensing / sparse proposal	Loihi 2 brief
Active Inference	"FEP is the control law"	Surprise → refuse prior	Friston FEP
Multi-agent	"Society of minds is safety"	Many propose, one commit	AutoGen; MetaGPT
Biology ~20 W	"We can't match it anyway"	Aspiration + accounting honesty	Sandberg; 2102.06273

4.2 ASCII map

	COMPETE	COMPOSE WITH WCA
	-----	-----
World models	"prediction is safety"	proposer / imagination
Etched/Groq/Taalas	"tokens/J is the whole game"	cheap proposal path
System One	"typed NL replaces commit law"	upstream decision tax cut
Lean agents	"proofs replace runtime gate"	offline properties
MCP/CUA	"tools are enough"	transport + irreversible class
CBF/shields	"already solved"	add cbf_ok conjunct

pH / Landauer	"theory without product"	EFA certificate language
Neuromorphic	"spikes replace LUTs"	sensing / sparse proposal
Active Inference	"FEP is the control law"	surprise → refuse prior
Multi-agent	"society of minds is safety"	many propose, one commit
Biology 20W	"we can't match it anyway"	aspiration + accounting honesty
Runtime cert monitors	"RV replaces product gate"	repair/monitor kinship

4.3 System One → WCA → plant (product diagram)

From SYSTEM_ONE_TO_WCA.md:

```
state → System One (typed branch) → only if machine action
  → Proposal (TLMM / agent u)
  → Commit Gate: lut ∧ safe [∧ cbf]
  → allow: plant_action = u | refuse: plant_action = 0 + RefuseReason[]
```

Deck line: *System One decides in software. WCA decides whether the machine is allowed to move.*

5. Energy / joule scoreboard (honesty tiers)

5.1 Measurement tiers (OpenIE discipline)

Tier	Name	Meaning	Allowed claim language
A	Surrogate	OpCounter × analytical E_* (joule.rs)	“surrogate joules,” never “board watts”
B	Post-synth estimate	Vendor cell libs / P&R power reports	“estimated post-synth W” with tool chain named
C	Board meter	Joulescope / shunt / MLPerf-class DUT	“measured W”

Current stack: **Tier A only**. board_synth_claimed=false always.

5.2 Scoreboard (verified numbers only)

System	Quantity	Value	Source	Notes
Human brain (classic)	Metabolic power order	~20 W	Sandberg https://arxiv.org/abs/1602.04019	Order-of-magnitude; not all compute.
Brain energy partitioning	Cortical ATP nuance	Computation ≪ total glucose power	https://arxiv.org/abs/2102.06273	Undercuts naive 20 W = AI budget.

System	Quantity	Value	Source	Notes
Landauer limit	Min heat / irreversible bit erase	kT ln 2	Landauer PDF	Floor, not CMOS design point.
CMOS op energy	pJ-scale ops by node	table values	Horowitz 2014	Calibrate E_* only.
DiffLogic	Throughput example	>1M MNIST img/s/CPU core [paper]	https://arxiv.org/abs/2210.08277	Energy depends on deployment.
DWN	FPGA energy/ latency	paper figures	https://arxiv.org/abs/2410.11112	Model for future DUT.
BitNet b1.58	Ternary LLM claim	match FP16 from ~3B class [paper]	https://arxiv.org/abs/2402.17764	Proposal kinship.
T-MAC	CPU table-lookup	~4x throughput / ~70% energy ↓ vs llama.cpp [paper]	https://arxiv.org/abs/2407.00088	Axis B.
TeLLMe	Edge FPGA ternary LLM	≤5–7 W class / tok/s in paper family	https://arxiv.org/html/2510.15926v2	Board-context gold standard.
NVIDIA H100	TDP	700 W class	NVIDIA datasheet [VENDOR]	Ceiling, not intelligence.
Groq / Etched / Taalas	tok/s , W	[VENDOR]	vendor sites	Exclude from scientific scoreboard.
Loihi 2	App-level W	workload-specific	Intel brief	Not drop-in LLM.
wca-lut-edge seed-1	Episode surrogate J	~8.6795e-10 ; committed 1 / refused 15	README / live run Sep 29 2026 EDT	[SURROGATE] ; board_synth_claimed=false
wca util/J	Internal scalar	commit-count utility	RESULTS / SOTA_SCAN	Easy to game; not MLPerf.

Scoreboard moral. Near-term Charlot metric: **task-quality / joule_tier_C** (board), with tier A labeled surrogate. Brain ~20 W is an existence proof, not a claim WCA can make.

5.3 Seed-1 golden (verified on box)

Live run (cargo run --release --bin wca-soc-loop -- --steps 16 --init-theta 1.5 --init-omega 3.0 --seed 1), Tue Sep 29, 2026 EDT:

- committed=1, refused=15, energy_only refuses=15
- J=8.679500e-10 **[SURROGATE]**
- ops: bram=64, lut_reads=16, lut6_toggles=3, ph_muls=64, ph_adds=48
- board_synth_claimed=false

6. Runtime nomenclature as product (wca.commit.v1)

6.1 Wire types

Type	Schema	Rust twin	Role
Proposal	Proposal.schema.json	schema.rs	Cheap propose; never self-commits
Certificate	Certificate.schema.json		LUT $\wedge$ energy ($\wedge$ optional CBF) + metrics
RefuseReason	RefuseReason.schema.json		lut_veto, energy_veto, cbf_veto, budget_exceeded, policy
CommitDecision	CommitDecision.schema.json		Envelope $\rightarrow$ plant_action

Compose enums: lut_and_energy | lut_and_safe | lut_and_safe_and_cbf | energy_only | lut_only.

6.2 Tao checklist applied to the schemas

Property	How wca.commit.v1 aims to satisfy it
Unambiguity	decision $\in$ {allow, refuse}; commit boolean tied to decision; refuse codes are typed, not free-form strings
Expressiveness	Covers LUT veto, energy veto, CBF veto, budget, policy; compose ablations for bake-offs
Suggestiveness	lut_allow $\wedge$ energy_ok reads like the engineering predicate
Natural transformation	MCP tools/call $\rightarrow$ Proposal $\rightarrow$ Certificate $\rightarrow$ CommitDecision is a natural map from agent tool space to plant space

6.3 MCP irreversible beachhead

From PRODUCT.md / mcp_gate_demo.json (verified --demo run):

- read_only may bypass certificate
- irreversible requires Certificate; refuse $\Rightarrow$ executed=false, executor never called
- default compose for gate demo: LutAndSafeAndCbf
- RecordingExecutor never touches real I/O in demo mode

7. Star Trek capability map (metaphor)

Fiction $\rightarrow$ engineering ownership. **No claim that WCA delivers Trek tech.**

Trek capability	Measure	Propose	Commit / Refuse	Human layer
Ship computer Q&A	Sensors + state	NL / System One	Usually N/A (read-only)	Trust calibration
Navigational course	Localization	Trajectory / world model	Energy + collision CBF + mission LUT	Helm override
Manipulator / tractor	Force sensing	Policy residual	PH passivity + wrench limits	Ethics / ROE
Holodeck safety	Occupant tracking	Scene generator	Hard safety interlocks = commit gate	Protocols-off = disaster argument <i>for</i> certificates
Replicator	Inventory / energy	Generative design	Mass/energy conservation certificate	Scarcity / law
Away-team agent	Multimodal sense	Hypothesis generator	Sample/destruction tools gated irreversible	Medical ethics
Multi-crew decision	Shared state	Multi-agent debate	Single commit authority	Captain / doctrine

Punchline. Star Trek already wrote refuse-to-commit as drama. Charlot's program: **safety protocols = executable notation with joule receipts.**

8. Whitepaper outline (ready to lift)

Working title: *Notational Intelligence $\rightarrow$ Commit Law: $LUT \wedge$ Energy Certificates for Agentic Physical AI*

1. **Introduction** — Brahe/Kepler framing; Lee cited generously; problem = irreversible commit under joule budgets.

2. **Related work** — NI lineage; DiffLogic/DWN/NeuralUT; TeLLMe/T-MAC; CBF/shields; pH-NN; RV certificate monitors; System One complement; MCP.
3. **Nomenclature** — `wca.commit.v1` schemas; compose predicates; RefuseReason taxonomy.
4. **Architecture** — $\text{sense} \rightarrow \text{propose}(\text{TLMMIagent}) \rightarrow \text{certify}(\text{LUT} \wedge \text{PH} [\wedge \text{CBF}]) \rightarrow \text{commit} \text{refuse}$; Rust hot path; Icarus oracle.
5. **Energy accounting** — OpCounter; tier A/B/C; `board_synth_claimed` discipline.
6. **Experiments (current software reference)** — seed-1 SoC golden; Safe PH false-allow; gate bake-off; lyap-plant dual compose; MCP gate demo; TLMM scale under joule budget.
7. **Experiments (required before strong claims)** — CBF baseline bake-off vs Ames-style QP; second plant; DUT joule tier C; $\text{util} \neq \text{commit-count}$.
8. **Limitations** — toy plants; surrogate J; no board synth; incomplete ISS / Bayesian barriers.
9. **Conclusion** — compose, don't compete; honesty as notational contribution.

Claims allowed now: compositional software-reference novelty; false-allow=0 Safe PH on toy grids; iso-correct TLMM; auditable OpCounter; MCP refuse-before-execute.

Claims forbidden now: board W; brain-matching efficiency; “first energy-aware AI”; beating Jev on latency; silicon leadership.

9. Demo / code appendix (real paths only)

9.1 Layout (from README)

```
wca-lut-edge/
  README.md
  ARCHITECTURE.md
  Cargo.toml           # Rust workspace (1.98 day-to-day)
  crates/wca-commit/   # SoC + burn + rtl-verify lib + bins
    src/
      schema.rs        # Proposal, Certificate, RefuseReason, CommitDecision
      lut.rs            # AllowLut
      ph.rs             # FixedPointPH / Safe
      cbf.rs / lyap.rs  # optional CBF / dual plant
      tlmm.rs           # ternary proposal path
      episode.rs        # SoC episode loop
      mcp_gate.rs       # irreversible tool gate
      joule.rs          # OpCounter × E_* [SURROGATE]
      bin/wca-soc-loop.rs
      bin/wca-mcp-gate.rs
      bin/wca-lut-burn.rs
      bin/wca-rtl-verify.rs
      bin/wca-lyap-plant.rs
      ...
  artifacts/schemas/wca.commit.v1/
  artifacts/mcp_gate_demo.json
```

artifacts/fpga/	# .v / .mem — Icarus sim OK; no board synth claimed
tools/py_train/	# DiffLogic train-only exception

9.2 How a reader runs demos

Verified toolchain: rustc 1.98.0 (2026-08-18).

```
cd /workspace/wca-lut-edge
rustc --version # expect 1.98.x

# Primary CI
cargo test -p wca-commit

# MCP Commit Gate demo (irreversible refuse / allow)
cargo run --release --bin wca-mcp-gate -- --demo --out artifacts/mcp_gate_demo.json

# SoC episode (seed-1 golden: committed=1 refused=15 J≈8.6795e-10)
cargo run --release --bin wca-soc-loop -- \
  --steps 16 --init-theta 1.5 --init-omega 3.0 --seed 1

# Safe PH mode
cargo run --release --bin wca-soc-loop -- --safe-ph --steps 16 --seed 1

# Dual compose lut ∧ safe ∧ cbf
cargo run --release --bin wca-lyap-plant -- --out artifacts/lyap_plant_report.json

# Burn allow LUT (+ optional PH / SoC wrappers)
cargo run --release --bin wca-lut-burn -- --allow-mask 21887 --n-bits 4 --with-ph --with-soc

# Bit-exact verify (requires iverilog + vvp)
cargo run --release --bin wca-rtl-verify -- \
  --verilog artifacts/fpga/wca_allow_lut.v --mem artifacts/fpga/wca_allow_lut.mem

# Schema roundtrip
cargo test -p wca-commit schema::
cargo test -p wca-commit mcp_gate::
```

9.3 Example CommitDecision (real file)

Source: artifacts/schemas/wca.commit.v1/examples/commit_decision_refuse.json — seed-1 step 0 pattern: LUT allows, energy vetoes, plant_action = [0.0], board_synth_claimed: false.

9.4 Crate map → diagram boxes

Diagram box	Code
TLMM propose	tlmm.rs, wca_tlmm*.v

Diagram box	Code
LUT allow	lut.rs, wca_allow_lut.{v,mem}
Energy / Safe	ph.rs, wca_ph_energy.v; --safe-ph
CBF / dual	cbf.rs, lyap.rs; wca-lyap-plant
Episode loop	episode.rs, wca-soc-loop
Wire types	schema.rs + schemas dir
MCP gate	mcp_gate.rs, wca-mcp-gate
Joules	joule.rs ([SURROGATE])

10. Implications for Medium / code / OpenIE

Action	Why
Keep wca.commit.v1 as the notation product	Moat = nomenclature, not weights
Ship MCP irreversible adapter	Beachhead where CUAs destroy state
Meter gate_J / proposal_J	Answers verification-energy counter
Optional cbf_ok conjunct	Closes Ames gap without abandoning LUT export
OpenIE / joule-lang kinship	Compile-time budgets [?] runtime certificates
Never flip board_synth_claimed without Vivado/Quartus + meter	Integrity
Medium #1 = Brahe→Kepler narrative; #2 = satiation	Public series; study = backbone

11. Bibliography (primary URLs only)

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Local stack

- README.md, ARCHITECTURE.md
- artifacts/SOTA_SCAN.md, PRODUCT_LANDSCAPE.md, PRODUCT.md, SYSTEM_ONE_TO_WCA.md, RESULTS.md
- artifacts/schemas/wca.commit.v1/
- artifacts/NOTATIONAL_INTELLIGENCE_RESEARCH.md (prior memo)
- artifacts/presentations/MEDIUM_SATIATION.md (Medium #2)

12. Honesty appendix

Item	Status
Lee essay quotes / paraphrase	Live fetch https://thesephist.com/posts/notation/ (2022-01-02)
Compile 2026 talk details	Secondary (site/YouTube listings); don't over-quote without transcript → residual [UNVERIFIED] on talk substance
TypeSafe Jev architecture / exact multiples	[VENDOR/PRESS] — see PRODUCT_LANDSCAPE
Groq / Etched / Taalas absolute tok/s and W	[VENDOR]
Brain = 20 W “compute”	Oversimplified; cite partitioning papers
wca episode joules	[SURROGATE] analytical OpCounter
board_synth_claimed	false
“First” compositional LUT [^] PH [^] TLMM product	Plausible white space per Sep 2026 scan—not a guarantee against unknown private systems
OpenIE joule-lang	Public GitHub; integration depth evolving , not peer-reviewed SOTA
Star Trek mapping	Metaphor only
Sapir–Whorf	Weak form only
LeCun 2022 AMI PDF	Prefer Meta blog; avoid unstable mirror hashes
Neural certificate runtime papers (2412.12996, 2507.11987, 2511.06341)	Fetches via search indexes in method window; PDF/HTML URLs listed—quote numbers only from primary text if used in future whitepaper

Item	Status
Exact McWhorter page cites beyond Pullum overview	Prefer Pullum / Webster critical discussion URLs already in memo

Method stamp: WebSearch + WebFetch + local artifact read + live cargo demos on 2026-09-29 EDT (continuing into Sep 30 window for study polish). No citations invented to fill holes.

13. Bottom line for David

1. **Keep Lee as Brahe**—cite him generously; do not punch down.
2. **Own Kepler**—`wca.commit.v1` + $LUT \wedge EFA$ as *runtime law*, with functional proof in this repo.
3. **Compose, don't compete** with System One, MCP, world models, Groq-class inference, CBF theory, and RV certificate monitors.
4. **Win the honesty war** on joules and certificates; that *is* the notational contribution.
5. **Ship the irreversible-tool beachhead** while the whitepaper strengthens plants, CBF baselines, and DUT meters.
6. **Public series:** Medium #1 (this Kepler frame) + Medium #2 (satiation) + this study as backbone.

End of study.